TAG-Guided Reasoning for Black-Box LLMs

Making Multi-Hop Reasoning Explicit

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Motivation

LLMs:

- Excel at knowledge-intensive tasks
- Strong performance with contextual understanding

Where they fail

- Struggle with **multi-hop reasoning**
- Depend on **implicit associations**
- Produce **hallucinated reasoning chains**

Core Problem

Reasoning is not **explicit**, **structured**, or **verifiable**

Limitations of Existing Methods

RAG:

- Retrieves text chunks independently
- No explicit relationships

Graph-based approaches:

- Require model-level access
- Shows how different entities are related

Gap: No structured reasoning for black-box LLMs

Our Approach: TAG Construction

Step 1: Build Structured Knowledge

- Convert text → **Text-Attributed Graph (TAG)**
- Extract **entities** (LLM)
- Extract **relations** (REBEL)

Graph Representation

- Nodes: entities + documents
- Edges: semantic relationships

Text → Graph

Our Approach: Graph-Guided Reasoning

Step 2: Graph-Based Retrieval

- Identify entities in the query
- Retrieve **multi-hop paths**
- Scoring mechanism for ranking paths

Step 3: Structured Prompting

- Encode subgraph as **JSON**
- Provide explicit reasoning paths to LLM

Key Idea

Reasoning is **externalized** → explicit and verifiable

Dataset: BioASQ 13b

Overview

- ∼5,389 biomedical questions curated by domain experts
- Each question is linked to supporting PubMed documents

Question Types

- Yes/No
- Factoid
- List
- Summary

Why BioASQ?

- Requires **multi-hop reasoning** over distributed evidence
- Contains rich **biomedical entities and relations**
- Grounded in **real scientific literature**

System Pipeline

Text Input

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Entity + Relation Extraction

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Heterogeneous Graph Construction

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Multi-hop Path Retrieval

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Ranking & Pruning

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Structured JSON → LLM Output

Graph Structure

Nodes

- Biomedical entities (drugs, proteins, diseases)
- Documents (PubMed abstracts with evidence)

Edges

- Entity–Entity: semantic relations (from REBEL)
- Entity–Document: `related-to` evidence links

Graph connects entities to evidence through explicit relations

Graph-Based Retrieval

1. **Query Processing**
   - Extract entities from question (LLM)
   - Handle missing entities via document fallback
2. **Multi-hop Path Retrieval**
   - Compute paths between entity pairs (BFS / Yen's)
   - Capture multiple reasoning chains
3. **Ranking**
   - Prefer informative relations (rarer edges)
   - Penalize hub nodes (high degree)
   - Rank documents using path support + query overlap

Structured Reasoning Injection

Subgraph Construction

- Select top-ranked documents
- Retain supporting multi-hop paths

Topological Encoding

- Convert subgraph into structured JSON
- Includes:
  - entities and documents
  - relations with costs
  - reasoning paths

LLM Integration

- Provide structured graph + question
- LLM synthesizes answer using explicit evidence

Experimental Setup

Compared Systems

- **Prompt-only**: LLM uses only question (parametric knowledge)
- **RAG**: BM25 retrieves top-k document chunks
- **TAG (ours)**: Graph-based multi-hop reasoning + structured input

Controlled Setting

- Same LLM (Gemma)
- Same prompts across all systems

Evaluation Metrics

Task-specific Metrics

- Yes/No → Macro-F1
- Factoid → MRR
- List → Entity-level F1

Summary Evaluation

- ROUGE-1, ROUGE-2, ROUGE-L

Purpose

- Task metrics → correctness
- ROUGE → answer quality & overlap

Performance Comparison

Model	List F1	Yes/No	Factoid
Prompt	0.186	**0.839**	0.221
RAG	0.137	0.568	0.205
TAG	**0.202**	0.774	**0.254**

Key Insight

- TAG improves **multi-hop tasks** (List, Factoid)
- Outperforms RAG across all metrics

Summary Evaluation (ROUGE)

Observations

- Prompt-only achieves highest **Recall & F1**
- TAG achieves highest **Precision**
- TAG consistently outperforms RAG

Interpretation

- TAG produces more **focused, precise answers**
- Prompt-only generates **broader but less grounded responses**

Key Observations

- TAG outperforms RAG on **all tasks**
- Comparable to prompt-only baseline overall

- Best improvements seen in:
  - List questions
  - Factoid questions

Indicates: Graph structure helps multi-hop reasoning

Why Does TAG Help?

- Provides **explicit relational structure**
- Connects evidence across documents
- Reduces reliance on implicit reasoning

Why RAG underperforms?

- Retrieves **isolated text chunks**
- Lacks explicit connections between evidence

Why Prompt-only still strong?

- Uses parametric knowledge for simpler tasks

Limitations

- Performance depends on **entity & relation extraction quality**
- Errors propagate through graph construction
- Additional **computational overhead**

- LLM may not fully utilize graph structure
- Graph still converted to text (JSON bottleneck)

Conclusion

- TAG improves **multi-hop reasoning** over RAG
- Achieves performance comparable to prompt-only baseline
- Provides **structured and interpretable reasoning**

Externalizing reasoning improves grounding

Future Work

- Improve relation extraction accuracy
- Better graph pruning and path selection
- Stronger evaluation of **faithfulness & grounding**

- Explore alternative graph encodings
- Study how LLMs utilize structured inputs